

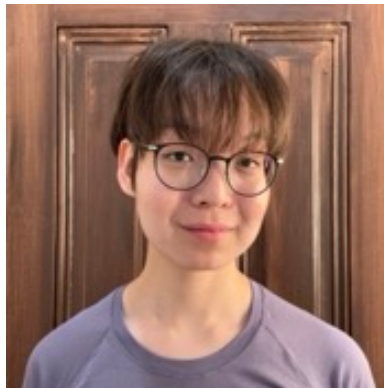
Statistics 151a: Linear models

UC Berkeley, Spring 2026

Instructors



Instructor: Ryan Giordano Office hours: Thur 11am-12pm
(389 Evans) Tues 4pm-5pm (on [Zoom](#)) rgiordano@berkeley.edu pronouns: He / him Office:
389 Evans Hall



GSI: Danat Dusingbekov Office hours: Wed 1pm-3pm, Mon
12pm-2pm (both in 428 Evans Hall) danat@berkeley.edu pronouns: He / him

! Important

Please reach out to [Danat](#) if you need to be manually added to BCourses.

Schedule

Lectures will be held on Monday, Wednesday, and Friday from 9–10am in Davis 534. The first lecture will be on Jan 21st, and the last will be on Friday May 1st.

Labs will be held on Wednesdays from 11 am – 1 pm and 3–5 pm in Evans 340.

([Link to official course calendar.](#))

Materials

Unless otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- [SMT: Statistical Models: Theory and Practice](#) Freedman
- [LME: Linear Models and Extensions](#) Ding
- [ISL: An Introduction to Statistical Learning](#) James, Witten, Hastie, Tibshirani
- [VDS: Veridical Data Science](#) Yu, Barter
- [ETM: Econometric Theory and Methods](#) Davidson, MacKinnon

Other books of interest to the class are

- [FPP: Statistics](#) Freedman, Pisani, Purves
- [RDS: R for Data Science](#), Wickham, Grolemund
- [ROS: Regression and other Stories](#) Gelman, Hill, Vehtari

Linear algebra review materials

Basic linear algebra is a serious prerequisite for this course. You can find a summary of useful review materials [here](#).

R programming review materials

This course will be conducting in the R programming language, and basic proficiency is assumed. Here are some useful review materials:

- [Berkeley SCF R bootcamp materials](#)
- [Berkeley D-Lab R fundamentals workshop](#)

Course Calendar

The following schedule is subject to change.

Table 1: Calendar (tentative)

Date	Day	Note	Unit	Topic	Assignment
Jan 21	Wednesday	Lecture	Introduction	Class policies	Lab 0
Jan 23	Friday	Lecture		Real-world questions	
Jan 26	Monday	Lecture		Inference and probability	
Jan 28	Wednesday	Lecture	Unit 1: Multilinear regression	Prediction and uncertainty	Lab 1
Jan 30	Friday	Lecture		Linear algebra review	HW 0
Feb 2	Monday	Lecture		Simple linear regression	
Feb 4	Wednesday	Lecture		Multilinear regression	Lab (review)
Feb 6	Friday	Lecture		———	Quiz 0
Feb 9	Monday	Lecture		Paths to the least squares objective	Lab 2
Feb 11	Wednesday	Lecture	Unit 2: Testing and inference	Transforming the regressors	
Feb 13	Friday	Lecture		Transforming the response	
Feb 16	Monday	Administrative holiday			
Feb 18	Wednesday	Lecture		Confidence intervals review	Lab 3
Feb 20	Friday	Lecture		The multivariate normal	HW 1
Feb 23	Monday	Lecture		Z and T tests	
Feb 25	Wednesday	Lecture		Model selection and F-tests	Lab (review)

Date	Day	Note	Unit	Topic	Assignment
Feb 27	Friday	Lecture		—	Quiz 1
Mar 2	Monday	Lecture		Asymptotics and consistency	
Mar 4	Wednesday	Lecture		Heteroskedasticity and grouping	Lab 4
Mar 6	Friday	Lecture	Unit 3: Regularization and machine learning	The bootstrap	
Mar 9	Monday	Lecture		Splines and basis expansions	
Mar 11	Wednesday	Lecture		Ridge regression	Lab 5
Mar 13	Friday	Lecture		Lasso regression	HW 2
Mar 16	Monday	Lecture		Cross validation and model selection	
Mar 18	Wednesday	Lecture		Information criteria (AIC and BIC)	Lab (review)
Mar 20	Friday	Lecture		—	Quiz 2
Mar 23	Monday	Spring break		—	
Mar 24	Tuesday	Spring break		—	
Mar 25	Wednesday	Spring break		—	
Mar 26	Thursday	Spring break		—	
Mar 27	Friday	Spring break		—	
Mar 30	Monday	Lecture	Unit 4: How regression can go wrong	Outliers and leverage	
Apr 1	Wednesday	Lecture		The influence function	Lab 6
Apr 3	Friday	Lecture		Regression to the mean	HW 3
Apr 6	Monday	Lecture		The FWL theorem	
Apr 8	Wednesday	Lecture		Omitted variable bias	Lab (review)
Apr 10	Friday	Lecture		—	Quiz 3
Apr 13	Monday	Lecture	Unit 5: Generalizations	Logistic regression	
Apr 15	Wednesday	Lecture		Poisson regression	Lab 7
Apr 17	Friday	Lecture		Nonlinear least squares	HW 4

Date	Day	Note	Unit	Topic	Assignment
Apr 20	Monday	Lecture		Random effects models	
Apr 22	Wednesday	Lecture		Hierarchical modeling	Lab (review)
Apr 24	Friday	Lecture		Hierarchical modeling	Quiz 4
Apr 27	Monday	Lecture	Unit 6: Special topic	TBD	
Apr 29	Wednesday	Lecture		TBD	Lab (projects)
May 1	Friday	Lecture		TBD	HW 5
May 4	Monday	Lecture	Final projects		
May 6	Wednesday	Lecture			Lab (projects)
May 8	Friday	Lecture			

Unless otherwise noted, the primary materials for the course are the lecture notes, which will be posted to the course website in advance of class. The following textbooks are useful supplementary texts and are all either freely available online or available electronically through the UC Berkeley library:

- [SMT: Statistical Models: Theory and Practice](#) Freedman
- [LME: Linear Models and Extensions](#) Ding
- [ISL: An Introduction to Statistical Learning](#) James, Witten, Hastie, Tibshirani
- [VDS: Veridical Data Science](#) Yu, Barter
- [ETM: Econometric Theory and Methods](#) Davidson, MacKinnon

Other books of interest to the class are

- [FPP: Statistics](#) Freedman, Pisani, Purves
- [RDS: R for Data Science](#), Wickham, Golemund
- [ROS: Regression and other Stories](#) Gelman, Hill, Vehtari